India Smart Utility Week 2026

Power Foundation of India

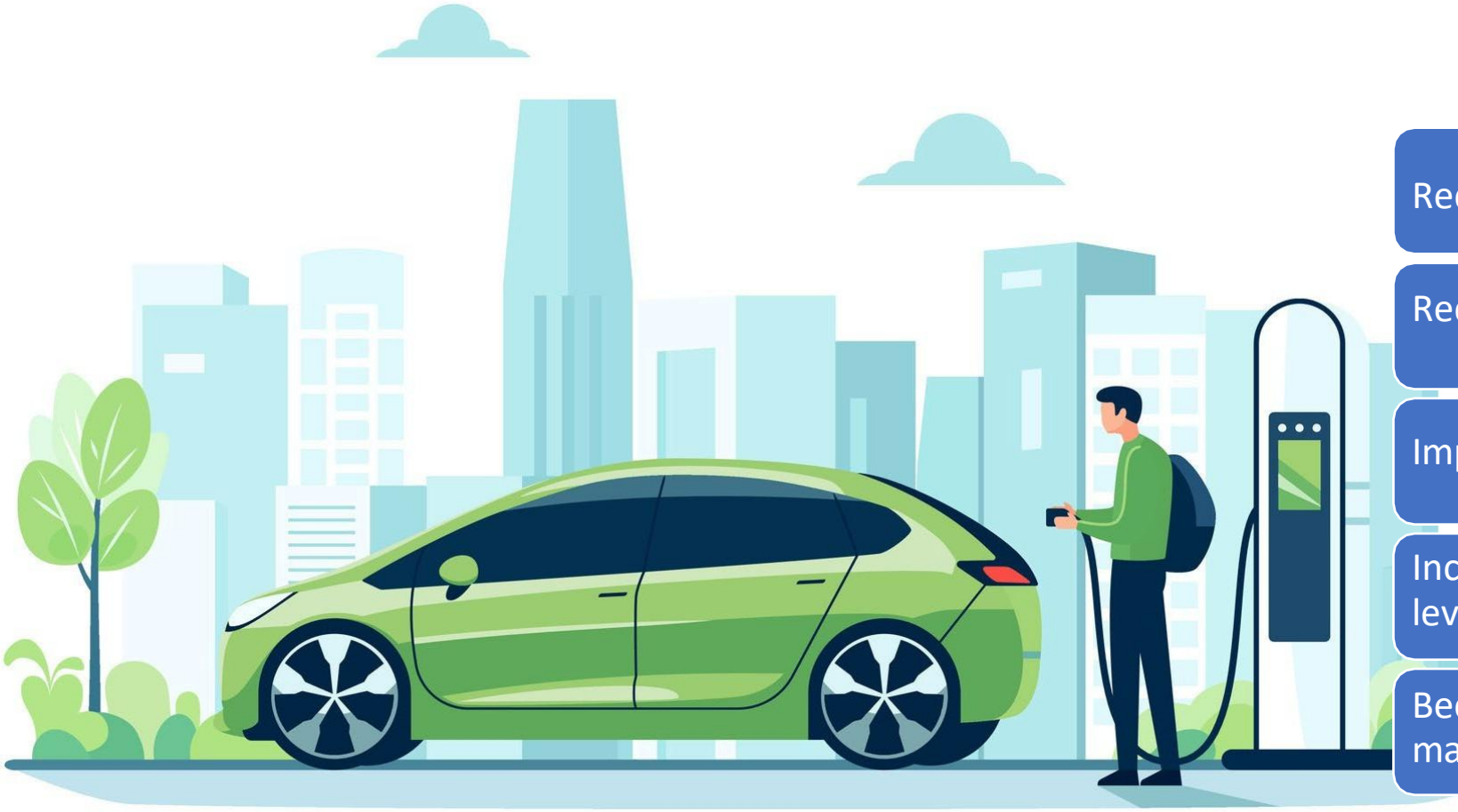
Knowledge Partner

ELECTRIC VEHICLES – A $200 BILLION OPPORTUNITY IN INDIA TO BE UNLOCKED

13th March, 2026

Power Foundation of India

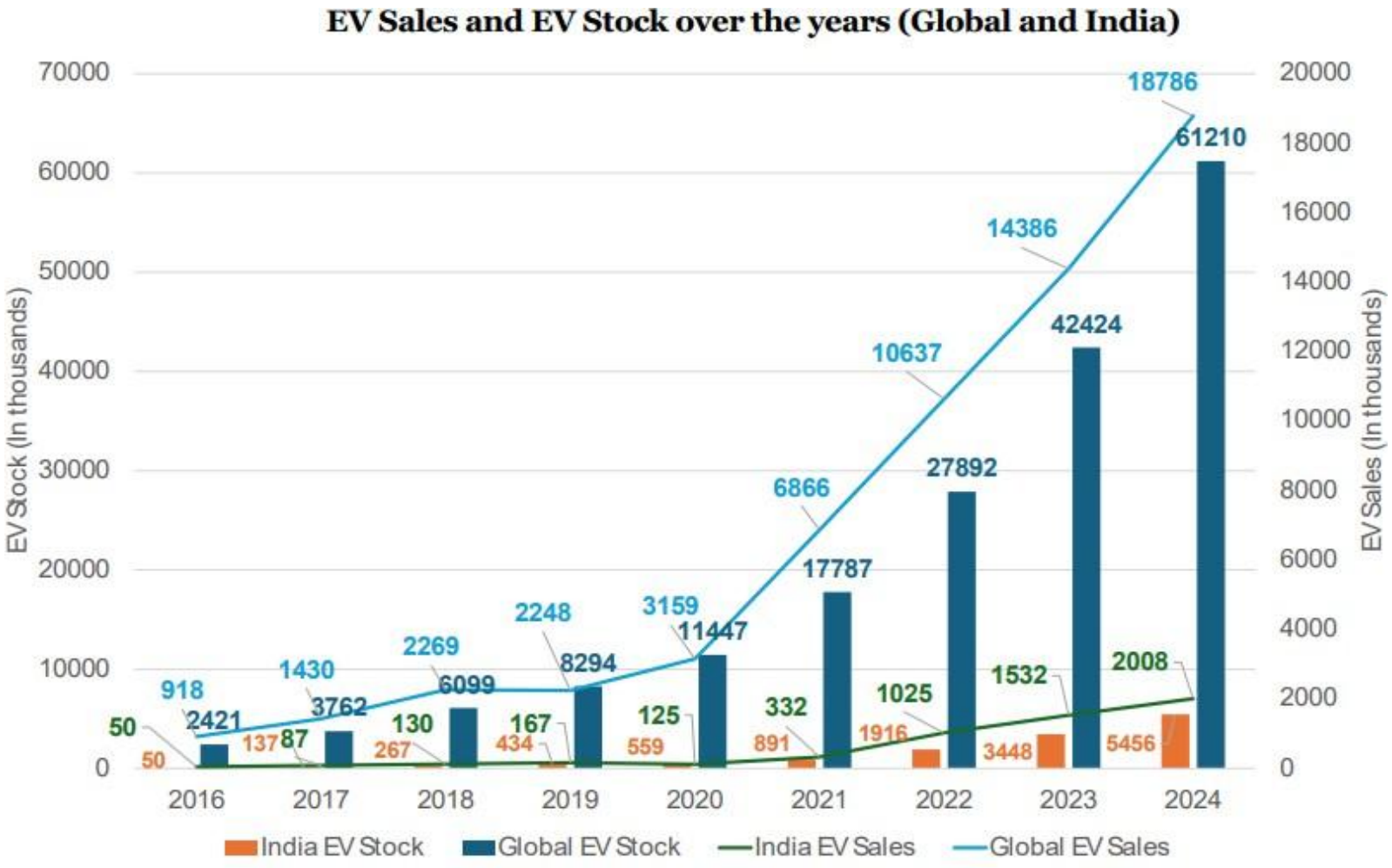
Why EV Transition?



- Reduce dependence on imported fuel
- Reduce Greenhouse Gases (GHG) emissions
- Improve air quality
- Increase share of renewable energy by leveraging the storage capacity of EV batteries
- Becoming a leader in a rapidly growing global market

INTRODUCTION |NITI Aayog Report

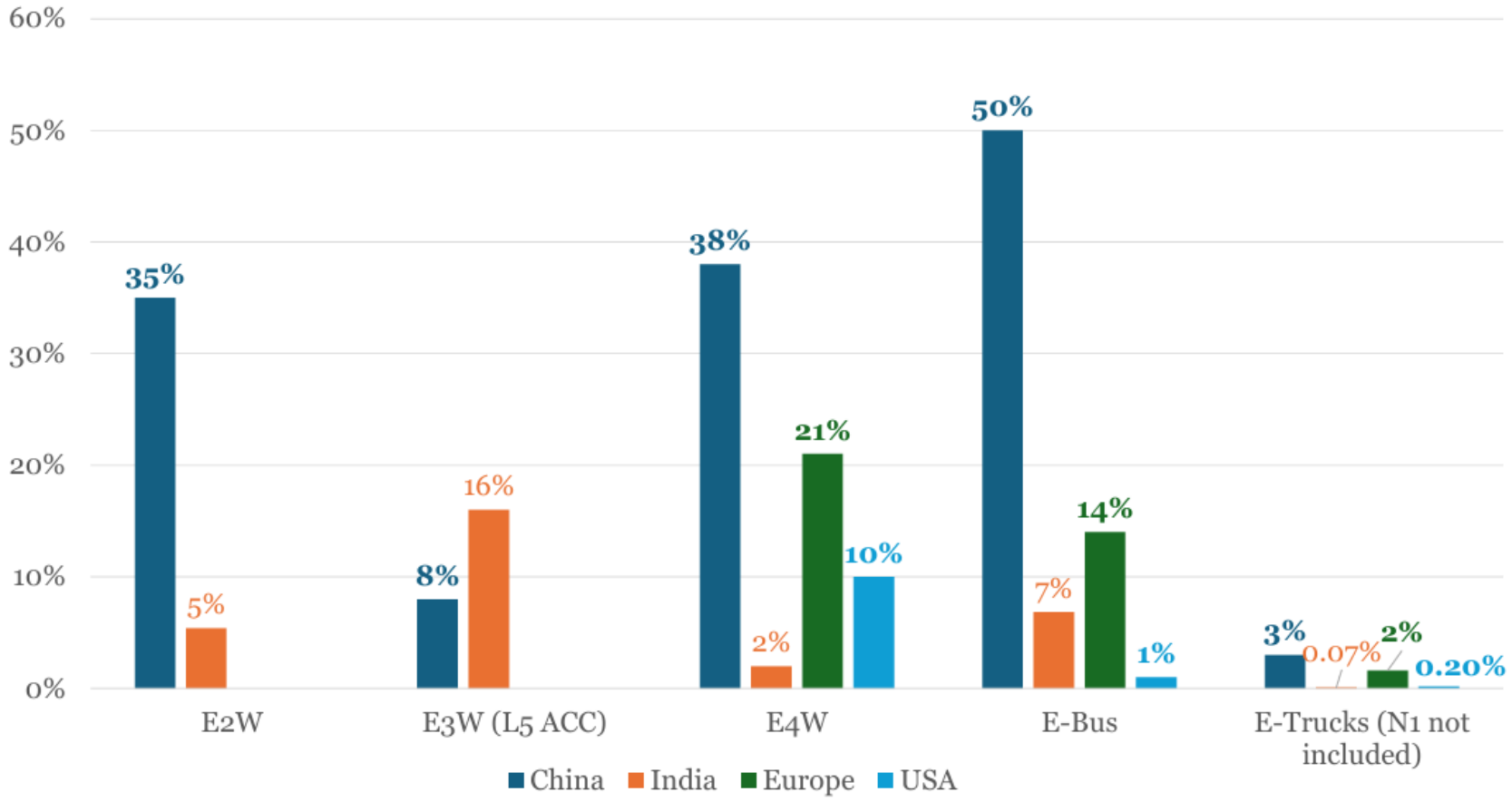
India's EV transition has been slow to start, but is picking up



Source: 1) IEA Global EV Data Explorer for global data, 2) VAHAN Portal for India data

INTRODUCTION |NITI Aayog Report

EV penetration rate across China, India, Europe & US in different vehicle segments in 2024



Source: 1) IEA Global EV Data Explorer for international data, 2) VAHAN Portal for India data

KEY CHALLENGES |NITI Aayog Report

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Financing Hurdles

Restricts fleet electrification (especially e-bus and e-truck)

Charging Inadequacy

Inadequacy of charging facilities and low utilization of existing public charging facilities limits EV usability

Lack of Awareness

Lack of adequate awareness and misconception hinder adoption

Data Gaps

Impedes informed decisions

BUSINESS MODEL FOR E-Bus

- **Conventional bus Cost – Rs 60 lakh**
- **Electric buses cost – Rs 1.2 crore**
  (50:50 :: Bus : Battery)

- **DTC: Rs 43/km**
- **Haryana : Rs 38.50/km**
- **Rajasthan: Rs 39.75/km**

- **Charging cost: Rs. 6 per kWh**
- **Battery Size: 350 kWh**
- **Range: 200 km**

Battery Leasing Model

Rs. 8.22 + **Rs. 8.22** + **Rs. 10.50** = **Rs. 26.94**

E-bus cost (per km) + Battery cost (per km) + Battery charging cost (per km) = **Total cost for e-Bus (per km)**

19.53 Lakh Total Number of Buses in 2030

5.85 Lakh e-Buses (Share of EV 30%)

Opportunity in E-Bus

Rs. 7 Lakh Cr.
Or
79 billion USD

BUSINESS MODEL FOR E-Truck

- **Conventional Truck Cost – Rs 11.90 lakh**
- **Electric Truck cost – Rs 30 lakh**

- **Charging cost: Rs. 6 per kWh**
- **Battery Size: 64.5 kWh**
- **Range: 162 km**

Running Cost of Conventional Truck

Mileage 11 kmpl

Distance 146,000 km

Rs. 7.39 per km

Rs. 3.63 per km

Running Cost of EV Truck

Battery Cost Rs. 1.24 per km

Running Cost Rs. 2.39 per km

1.08 Crore
Total Number of Trucks in 2030

32.4 Lakh Trucks
(Share of EV 30%)

Opportunity in E-Truck

Rs. 9.72 Lac Cr.
Or
109 billion USD

CHALLENGES FOR THE POWER SECTOR

India's EV market is **projected to grow to a $200 billion opportunity** by 2030, with significant penetration in public transport and private mobility.

Uncoordinated EV charging, particularly during evening hours, can amplify system **peak demand**, stressing grid infrastructure.

Increased short-term power procurement during peak periods may raise DISCOMs' **Average Power Purchase Cost (APPC)** and overall cost of supply.

Smart tariffs, including **Time-of-Day (ToD)** and **real-time pricing**, can promote off-peak charging, reduce grid stress, and improve cost efficiency.

It is important that India achieves its EV transition goals while ensuring that the **power sector remains cost-effective and resilient**

Study the incidence of Electric Bus Charging on DISCOMs' Network and Scope for Optimal Power Purchase

(PFI Report)

March 2026

Supported By

BSES Rajdhani Power Limited

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BSES Yamuna Power Limited

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(Autonomous Society under the aegis of Ministry of Power, Government of India)

Acknowledgements Key Contributors

Acknowledgements

Amal Sinha, Group CEO, BSES Delhi DISCOMs

BSES is committed to powering a greener, smarter, and more resilient energy future for Delhi, ultimately benefiting consumers with reliable, cost-effective, and environmentally responsible energy solutions. In this direction, this Report of PFI will certainly support DISCOMs in achieving its target.

Abhishek Ranjan, CEO, BSES Rajdhani Power Ltd.

PFI has done commendable work of addressing the core issue of optimisation of Peak Demand of DISCOMs through collaboration with BRPL, which will be fruitful for the consumers of Delhi at large.

Dwijadas Basak, CEO, TATA Power Delhi Distribution Ltd.

TPDDL, in its continuous efforts of give impetus to clean and green technology initiatives, is leading as a front-runner utility in the country by scaling up its electric vehicle fleet. PFI Report will definitely help in optimising the Charging of the Electric Fleet.

Brajesh Kumar, COO, BSES Yamuna Power Ltd.

PFI has actively engaged with Delhi DISCOMs and has come out with very pertinent recommendations for managing peak demand in Delhi. The study findings will help DISCOMs and policymakers in creating a sustainable and resilient power infrastructure ecosystem for the benefit of our esteemed customers.

Anshuman Srivastava Executive Director, Power Foundation of India

I commend the efforts of the PFI for producing this comprehensive and forward-looking analysis. The insights from this report will be valuable for all stakeholders as we work together to deliver on India's energy aspirations.

E-mobility is the need of the hour, but it is struggling with core issues, such as charging at Peak hours at a costlier rate and non-rationalised tariffs for charging EVs, etc. This report makes a timely and valuable contribution by addressing these very questions in a structured and data-driven manner.

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Contents

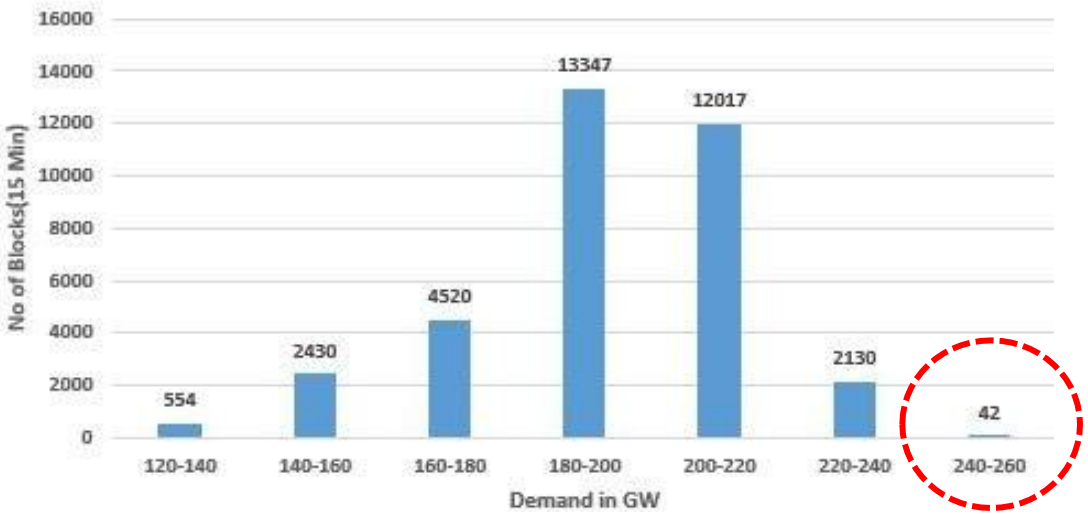
- Problem Statement
- Objective of the Study
- Approach and Methodology
- Analysis
- Findings
- Optimization Measures- Shifting of Load, BESS C deferral of Network (avoided Peak)
- Recommendations to Stakeholders
- Conclusion

Problem Statement |Peak Demand all India

PEAK DEMAND, THE MENACE

- Peak demand, the highest average load measured in kVA or kW at the point of supply of a consumer during a consecutive period of 15/30 mins during a billing cycle
- All India Peak Demand increased from **157 GW in FY 2015-16 to 250 GW** in FY 2024-25 (*CEA*), however, it occurs for only ~0.1% of the total time in a Year (in 42-time blocks), Fig 1.
- DISCOMs, as mandated u/s Sec. **43 of the Electricity Act, 2003, for "Duty to Supply on request"** required to maintain the network to meet Peak Demand and incur commensurate Power Purchase Cost C Network Cost to meet such Peak Demand.
- **Supreme Court in Judgement dtd. 1G/05/2023** in KC Ninan v. Kerala State Electricity Board has ruled that "*the owner or occupier has the statutory right to "demand" electricity for the premises under their use or occupation.*"

Fig 1: Frequency Distribution of Demand in FY 2024-25
(Grid Controller of India)



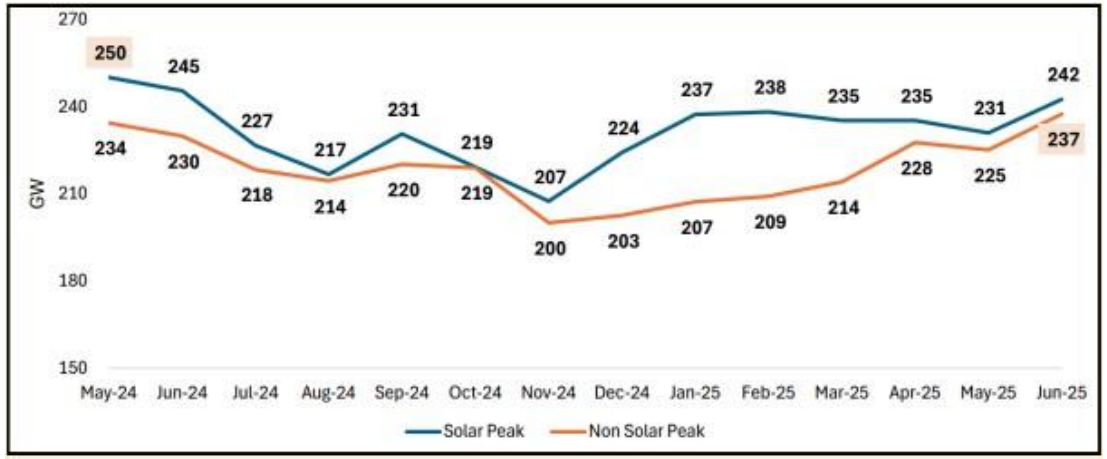


Fig 2: Peak Demand Monthly Trend in FY 2024-25, *Source: GRID India*

Huge Solar Power meets the Peak Demand during daytime, but vanishes during night peak

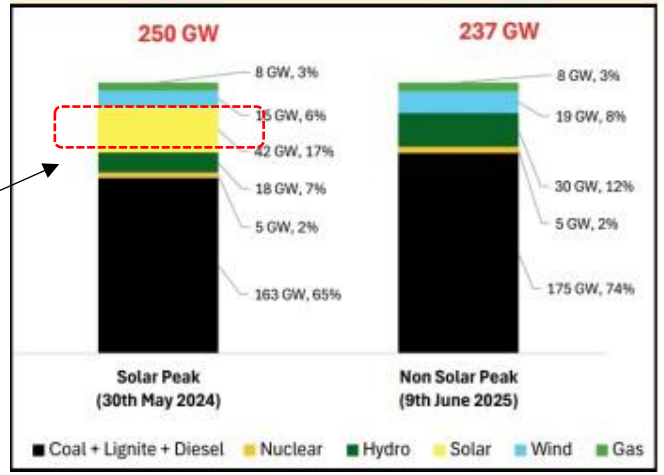


Fig 3: Source-wise contribution in Peak Demand FY 2024-25, *Source: GRID India*

Problem Statement |Peak Demand Delhi DISCOMs

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Peak load FY 2023-24

Delhi: 7438 MW
(~1% of the time)

- BRPL: 3758 MW
- TPDDL: 2218 MW
- BYPL: 1800 MW

Highly dense and populated DISCOMs, with consumer density of ~4,000/ sq km for BRPL G TPDDL and G,000/ sq km for BYPL

Fig 4. Delhi Load Duration Curve FY 2023-24 *(Delhi SLDC)*

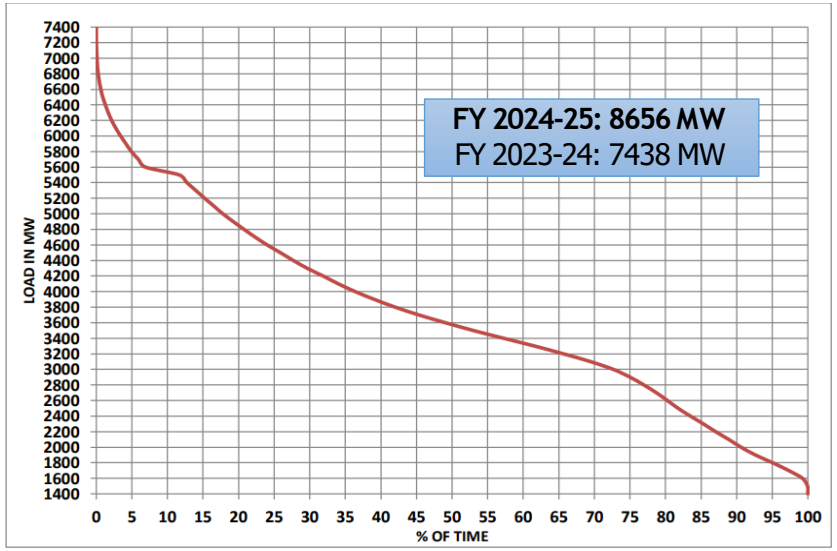
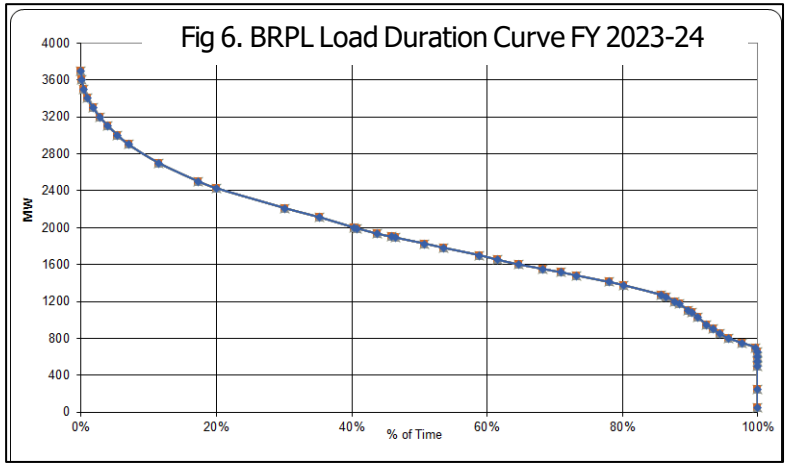
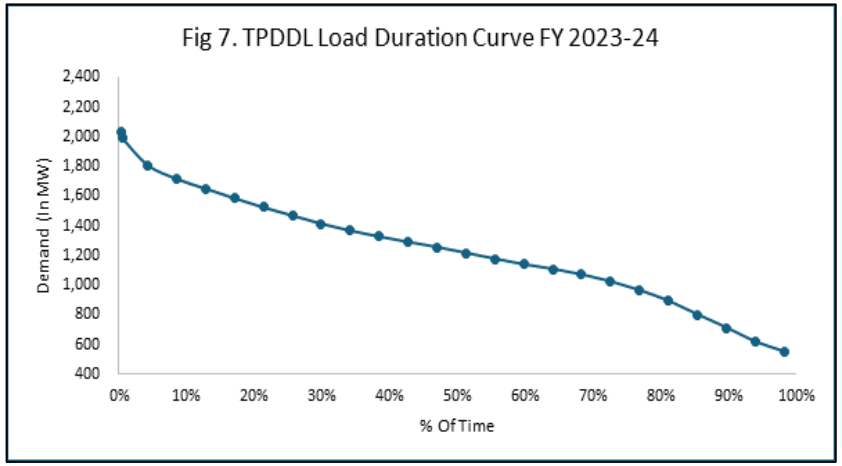


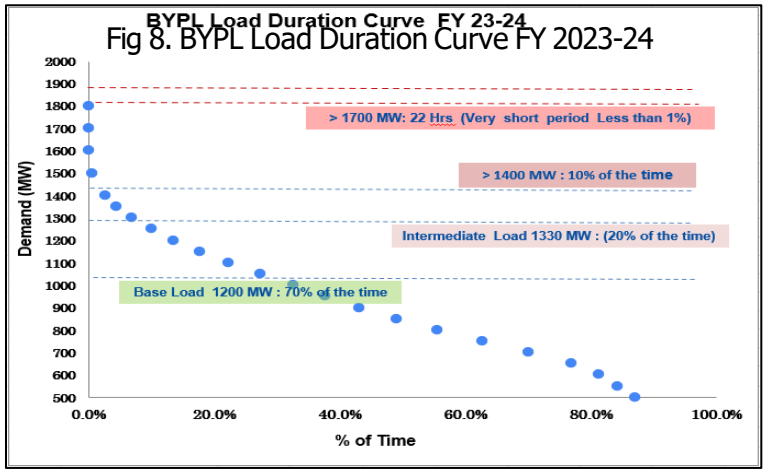
Fig 5. Frequency Distribution of Demand in FY 2023-24, Delhi

~1,000 MW of differential load (Δ) occurred for just 1% of the time in a year, **for 113 hours only**

LOAD REMAINED ABOVE IN MW	DURATION IN HOURS	(%) OF TIME
7400	0.00	0.00
7300	0.50	0.00
7200	1.50	0.01
7100	5.50	0.04
7000	11.50	0.09
6900	19.25	0.15
6800	33.25	0.27
6700	53.50	0.43
6600	79.25	0.64
6500	112.75	0.91
6400	159.75	1.28
6300	209.50	1.69
6200	269.00	2.16
6100	341.50	2.75
6000	426.00	3.43
5900	518.25	4.17
5800	613.50	4.93
5700	736.00	5.92







Problem Statement |Peak Demand Delhi DISCOMs

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Fig G. Delhi Peak Demand FY 2024-25 G 2023-24 *(Delhi SLDC)*

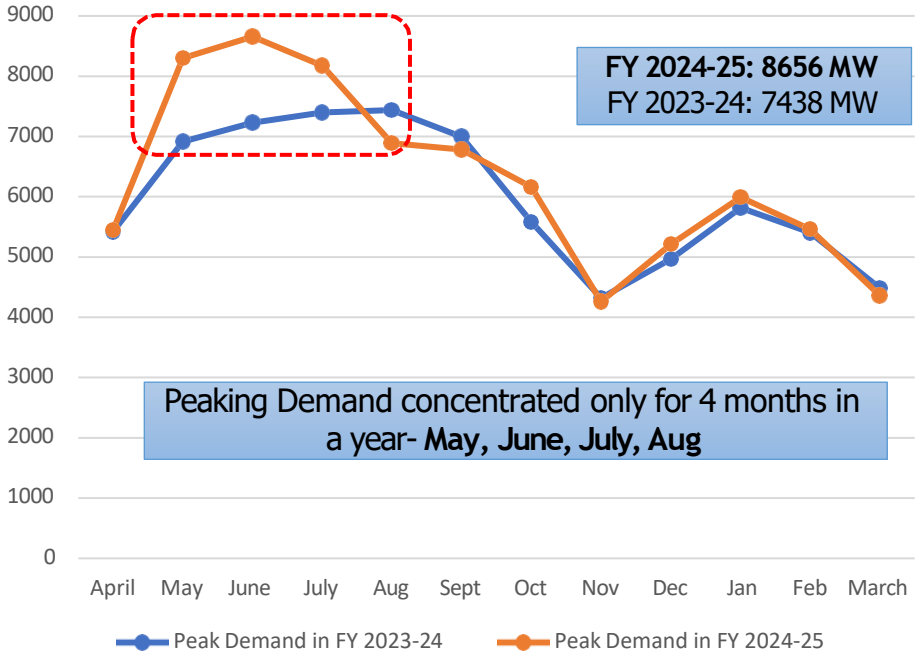

Fig 10. Delhi Block wise Peak Demand in FY 2024-25

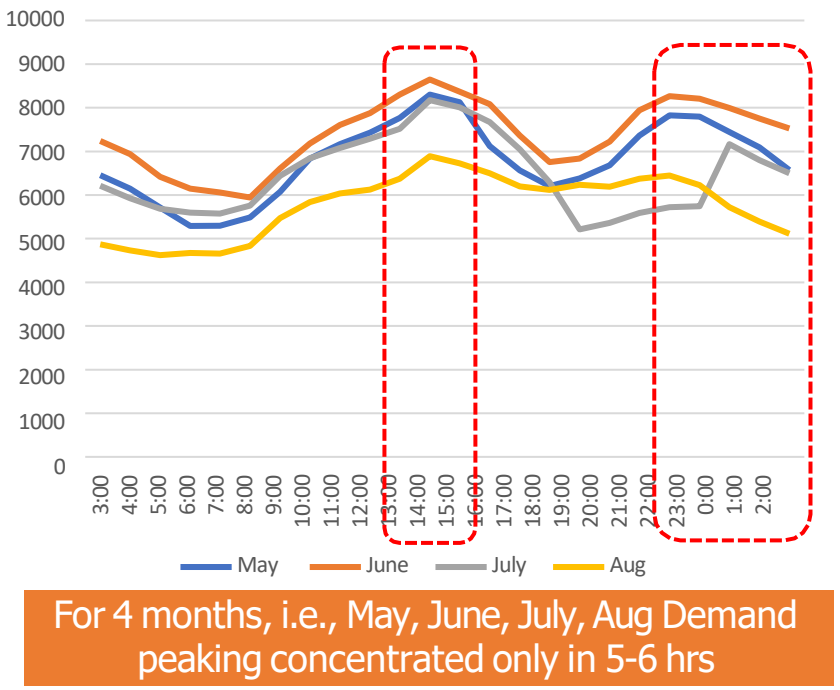

Fig 11. Short Term rate in Peak hours (Rs./kWh)

Time Block	BRPL	TPDDL	BYPL
June 2024			
2100-2130	9.67	10	10
2130-2200	9.62	10	10
2200-2230	9.65	10	10
2230-2300	9.76	10	10
2300-2330	9.77	10	10
2330-2400	9.76	10	10
0000-0030	9.88	9.84	10
0030-0100	9.83	9.5	10
July 2024			
2100-2130	9.67	10	10
2130-2200	9.62	10	10
2200-2230	9.65	10	10
2230-2300	9.76	10	10
2300-2330	9.77	10	10
2330-2400	9.76	10	10
0000-0030	9.88	10	9.58
0030-0100	9.83	9.52	8.51

- DISCOMs coincident Peak Demand translates to States' Coincident and majority of States' coincident translates to All India coincident.
- **Differential (Δ) Peak load is ~1000 MW occurred for ~1% of the time, i.e., around 113 hours in FY 2023-24**
- To meet Peak load, Delhi DISCOMs **procures Short term power @ avg rate of Rs. 8.5/kWh, ~ Rs. 3000 Cr in FY 2024-25** (~10% of Total Power Purchase Quantum). Power Purchase for 4 months- May, June, July, Aug is ~**Rs. 1400 Cr** out of which Power Purchase in peaking 5-6 hrs, is ~**Rs. 750 Cr.**
- To meet Δ Peak load of ~1000 MW, Delhi DISCOMs **procures Short term power @ Rs. 10/kWh (Fig. 11).** At National level, the power purchase cost required to meet the Peak Load would be staggering

Problem Statement |Peak Demand Delhi + E-Bus

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Jun-24	Peak Demand		
Time Block	BRPL	TPDDL	BYPL
0000-0030	2.44%	2.38%	2.42%
0030-0100	2.39%	2.34%	2.38%
0100-0130	2.34%	2.28%	2.33%
0130-0200	2.29%	2.23%	2.28%
0200-0230	2.23%	2.18%	2.23%
0230-0300	2.18%	2.13%	2.17%
0300-0330	2.12%	2.07%	2.11%
0330-0400	2.07%	2.02%	2.06%
0400-0430	2.01%	1.96%	1.99%
0430-0500	1.96%	1.91%	1.94%
0500-0530	1.90%	1.84%	1.88%
0530-0600	1.82%	1.78%	1.83%
0600-0630	1.75%	1.72%	1.79%
0630-0700	1.71%	1.68%	1.77%
0700-0730	1.67%	1.65%	1.74%
0730-0800	1.65%	1.64%	1.71%
0800-0830	1.64%	1.63%	1.67%
0830-0900	1.65%	1.63%	1.66%
0900-0930	1.70%	1.71%	1.70%
0930-1000	1.78%	1.85%	1.78%
1000-1030	1.85%	1.94%	1.84%
1030-1100	1.91%	2.02%	1.92%
1100-1130	1.97%	2.08%	1.98%
1130-1200	2.01%	2.13%	2.04%
1200-1230	2.06%	2.19%	2.10%
1230-1300	2.11%	2.23%	2.14%
1300-1330	2.14%	2.16%	2.16%
1330-1400	2.20%	2.25%	2.20%
1400-1430	2.27%	2.34%	2.25%
1430-1500	2.33%	2.40%	2.29%
1500-1530	2.35%	2.43%	2.32%
1530-1600	2.34%	2.42%	2.35%
1600-1630	2.32%	2.38%	2.34%
1630-1700	2.26%	2.33%	2.29%
1700-1730	2.20%	2.25%	2.23%
1730-1800	2.11%	2.15%	2.17%
1800-1830	2.02%	2.07%	2.10%
1830-1900	1.96%	2.01%	2.04%
1900-1930	1.94%	1.98%	1.99%
1930-2000	1.98%	1.99%	1.99%
2000-2030	2.04%	2.01%	2.01%
2030-2100	2.11%	2.03%	2.05%
2100-2130	2.20%	2.05%	2.09%
2130-2200	2.29%	2.16%	2.18%
2200-2230	2.38%	2.26%	2.28%
2230-2300	2.45%	2.34%	2.36%
2300-2330	2.47%	2.38%	2.42%
2330-2400	2.47%	2.39%	2.44%

Fig 12. E-Bus Charging during Peak hours in Delhi

June-24	Peak Demand			E-Bus Charging		
Time Block	BRPL	TPDDL	BYPL	BRPL	TPDDL	BYPL
1300-1330	2.14%	2.16%	2.16%	3.20%	3.69%	3.65%
1330-1400	2.20%	2.25%	2.20%	3.40%	3.88%	3.73%
1400-1430	2.27%	2.34%	2.25%	3.37%	3.95%	3.73%
1430-1500	2.33%	2.40%	2.29%	3.52%	4.00%	3.75%
1500-1530	2.35%	2.43%	2.32%	3.49%	4.03%	3.98%
1530-1600	2.34%	2.42%	2.35%	3.51%	3.90%	3.61%
1600-1630	2.32%	2.38%	2.34%	2.93%	3.71%	2.85%
2100-2130	2.20%	2.05%	2.09%	1.99%	2.66%	3.33%
2130-2200	2.29%	2.16%	2.18%	3.07%	3.91%	4.06%
2200-2230	2.38%	2.26%	2.28%	3.85%	3.76%	3.62%
2230-2300	2.45%	2.34%	2.36%	3.42%	3.20%	3.51%
2300-2330	2.47%	2.38%	2.42%	2.76%	3.77%	4.10%
2330-2400	2.47%	2.39%	2.44%	3.11%	4.17%	4.17%
0000-0030	2.44%	2.38%	2.42%	3.53%	3.91%	3.43%
0030-0100	2.39%	2.34%	2.38%	3.71%	3.81%	2.90%
0100-0130	2.34%	2.28%	2.33%	3.27%	3.98%	3.30%
0130-0200	2.29%	2.23%	2.28%	3.00%	3.71%	3.52%

Load pertaining to Domestic, Commercial, Industrial C other consumers is not particular to peak timings unlike EV Charging which comes during Peak hours in Summers. **Impact !!**

E-BUS CATALYSING PEAK DEMAND PROBLEM

While operation of Electric buses compared to Oil and gas-based buses will reduce pollution but unless properly planned, growing penetration of EVs may have a destabilising effect on the grid, may hamper cost economics of the Distribution Companies and increase the Average Cost of Electricity Supply to Retail consumers.

EVs demand electricity for charging according to convenience, which may coincide with the Peak Demand, pushing it up resulting in purchase of Costlier Short-Term Power to meet the demand.

Problem Statement |Role of Surging Electric Fleet

SURGENCE OF ELECTRIC FLEET

Fig 13. Surgence of EV (% of the total)

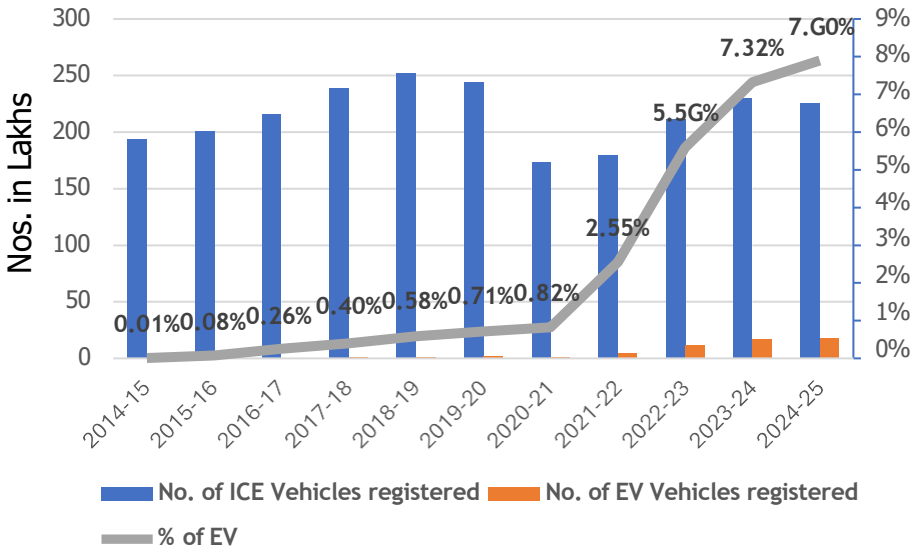
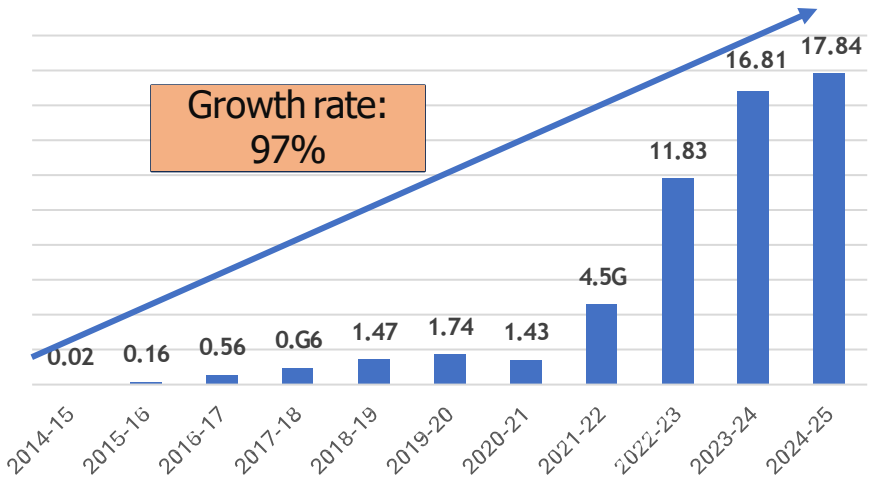


Fig 14. No. of EV Vehicles registered (in lacs)



Source: PIB, GoI

During last 10 yrs, EV registered CAGR of G7%.

During FY 2024-25, EV sold is 8% of the total vehicles.

Given the thrust on Green mobility, EV are expected to rise significantly in the coming time.

As a result, the Energy requirement will also rise considerably.

As per CEA (EPS), estimated EV on the road by FY 2031-32 would be **4G1 lakhs (14% of the total)**, power consumption of **27,000 MU out of total consumption of 24,73,736 MU at All-India level.**

Power Sector Report by OmniScience Capital, estimated EV on the road by FY 2034-35 would be 1,C20 lakhs, power consumption 15S,000 MU.

India's Target of 30% EV market share by 2030, **EV30@30**

EV30@30 INCREASING UPTAKE OF ELECTRIC VEHICLES
A CAMPAIGN OF THE CLEAN ENERGY MINISTERIAL

ELECTRIC BUS CHARGING- CATALYSING PEAK DEMAND MENACE

ABOUT E- BUSES

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ENERGY GUZZLER

E_BUS PARAMETERS 12 METRE *

- **Capacity of Battery: 268 kW**
- DC Fast Chargers of 240 kW each
- **Charging time: ~2 hrs (80% charge in 1 hr)**
- Range of E-Bus ~230 km
- Operational hours of E- Bus : 8 hr, Single trip
- Li-ion Battery with useful life of 8 years depending on SOC
- No. of charges available per E-Bus at the Depot : 1:5
- Charger Cost: 14~15 lakh

No. of E-bus	10,000	Nos
No. of plies in a day	2	Nos
Battery	268 (12-m) 200 (9-m)	kW
Charging Time	3	hrs
Consumption (annual)	**35G2**	**MU**

- 12-m E-Buses with 268 kW and 9-m Buses with 200 kW battery pack
- 80:20 ratio for 12-m and 9-m E-Buses

As on July 2024, 9143 nos of e-Bus, all India level
Source: *PIB, Govt. of India*

*For 12-m E-Buses
Source: Report on "THE ROAD AHEAD FOR PRIVATE ELECTRIC BUSES IN INDIA", Shakti Foundation & USAID

No of E-buses estimated to reach to 1,50,000 by 2030 (market Research)

Problem Statement |Role of Surging Electric Fleet

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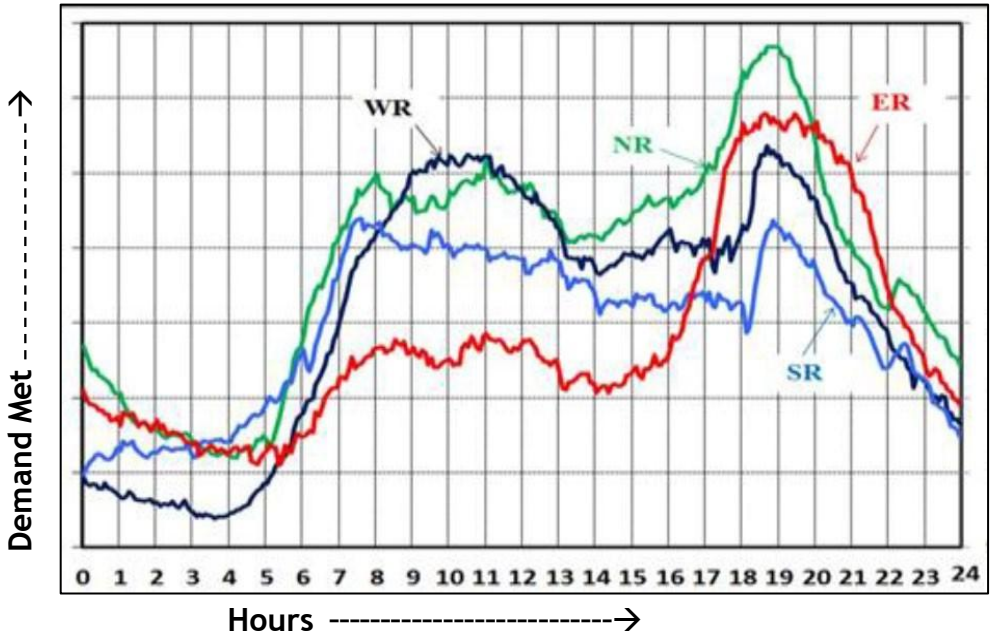
SURGENCE OF ELECTRIC FLEET |BURDEN ON PEAK DEMAND

- Road transportation accounts for 12% of CO2 emissions *(NITI Aayog)*. The GoI, with objective of reducing emissions from Transportation and improving the air quality, promoting adoption of Electric Vehicles (EVs). Government provides subsidies on purchasing EVs through FAME (Faster Adoption and Manufacturing of Electric Vehicles) India scheme.
- Other policy support programs like Production Linked Incentive (PLI) schemes for domestic vehicle manufacturing and battery cell production, Tax reduction on EV chargers and charging stations, and Capital subsidies to Oil marketing Companies (OMCs) for setting up public EV charging infrastructure.
- As on March'25, 2152 E-Buses in Delhi. 1752 E-buses under DTC and 400 E-buses under the DIMTS cluster scheme. This includes 12-meter Buses, 9-meter smaller Buses.
- E-Buses by virtue of their operations, Charge during night hours and ply during daytime but DISCOMs experience peak during nighttime, *(as shown in Graph)* which may coincide with the E-Bus Charging.



ELECTRIC BUS CHARGING- CATALYSING PEAK DEMAND MENACE

E- BUS DEPOTS

ENERGY GUZZLER

As per [Power Sector Report by OmniScience Capital]

- 35 lacs E-Buses estimated on road by FY 2034-35
- Power consumption for the same would be 33,000 MU.

	E-BUS DEPOT	DEPOT LOAD
BRPL	NEHRU PLACE	5 MW
	MAYAPURI	4 MW
	MUNDELA	5 MW
	SUKHDEV VIHAR	7 MW
BYPL	RAJGHAT	4 MW
	HASSANPUR- 1	8 MW
TPDDL	BARWALA	12 MW
	WAZIRPUR	7 MW
	SUBASH PALACE	8 MW
	ROHINI, SEC- 6	4 MW
	KINGSWAY CAMP	3 MW
	ROHINI, SEC- 16	4 MW
		65 MW*

* As on Nov, 2024

Capacity in FY 2025-26 : 268 MW
Capacity in FY 202G-30 : 500 MW

RULES ON EV TARIFF AND EV TARIFF SCHEDULE

Power Foundation of India

Ministry of Power notification dtd. 17/0G/2024 for Electric Vehicle Charging Infrastructure:

*"3. **Tariff for supply of electricity to EV charging stations***

(1)The tariff for supply of electricity to EV Charging Stations shall be single part and shall not exceed "ACoS" till 31.03.2028.

(2) The Distribution Licensee will charge 0.7 times the ACoS during solar hours (3:00 AM to 4:00 PM) and 1.3 times ACoS during non-solar hours (remaining hours).

(3) Each EV charging station must have separate metering arrangements to accurately record consumption and apply the appropriate tariff.

(4) Distribution Licensee may provide sub metering for EV charger, behind-the-meter of an existing HT connection."

- Costly ST Power Purchase
- No Fixed Charges
- Least Energy Charges Rs.4/kVAh
- ToD (Off-peak Rebate of 20%)

  4 AM to 10 AM

Clause 8.3 of Tariff Policy, 2016:

*"For achieving the objective that the **tariff progressively** **reflects the cost of supply of electricity**, the Appropriate Commission would notify a **roadmap such that tariffs are brought within** **±20% of the average cost of supply**. The road map would also have intermediate milestones, based on the approach of a gradual reduction in cross-subsidy"*

EV TARIFFS IN DIFFERENT STATES

Power Foundation of India

TARIFF SCHEDULE IN DELHI

Sr. No.	Category	Fixed Charges	Energy Charges
10. Charging Stations for E-Rickshaw/E-vehicle on Single Point Delivery/ Swapping of Batteries			
10.1	Supply at LT	-	4.50 Rs./kWh
10.2	Supply at HT	-	4.00 Rs./kVAh

Comparison of ABR of EV-CS with ACoS of DISCOMs (Rs./kWh)

DISCOMs	Average Cost of Supply (ACoS)	Average Billing Rate (ABR) for EV-CS	ABR (EV-CS) as a percentage of ACoS (%)	Tariff for EV-CS as per MoP Guidelines (Sept'24)	
				Solar Hours	Non-Solar Hours
BRPL	9.9	4.11	41%	6.93	12.87
BYPL	8.67	4.19	48%	6.07	11.28
TPDDL	9.55	4.52	47%	6.69	12.42
Total	**9.5**	**4.25**	**45%**	**6.65**	**12.35**

TARIFF SCHEDULE IN OTHER STATES

State	EV Tariff (FY 2025-26)	
	Fixed Charge (Rs./kVA/month)	Energy Charge (Rs./kVAh)
Maharashtra	Nil	9.25
Karnataka	200	4.50
Rajasthan	150	6.00
Uttar Pradesh	Nil	7.30
Gujarat	25	4.00
Telangana	100	6.00

- Unlike other States, Delhi has very less tariff for EV - Rs. 4/kVAh
- Very high Cross subsidization levels
- No Fixed Charges

E-BUS WORKING MODEL contd..

Power Foundation of India

Electric Bus Depots

Charging Infrastructure - Aggregators

TATA MOTORS | ASHOK LEYLAND | JBM Group

Land
Owned by DTC
Given on lease

Charging Stations

EXICOM, ABB, BHEL, TELUS

Tariff Rs. 4.00/kWh (HT)

ACS- Rs. 8.33/kWh

DISCOMs

TATA POWER-DDL | BSES

ABR- Rs. 8.90/kWh

E-Bus

Fixed Charge of Rs. 43/km to Aggregators

DTC

Delhi Transport Corporation

Charged fare for using E-Bus Service

Consumers of E-Bus Service

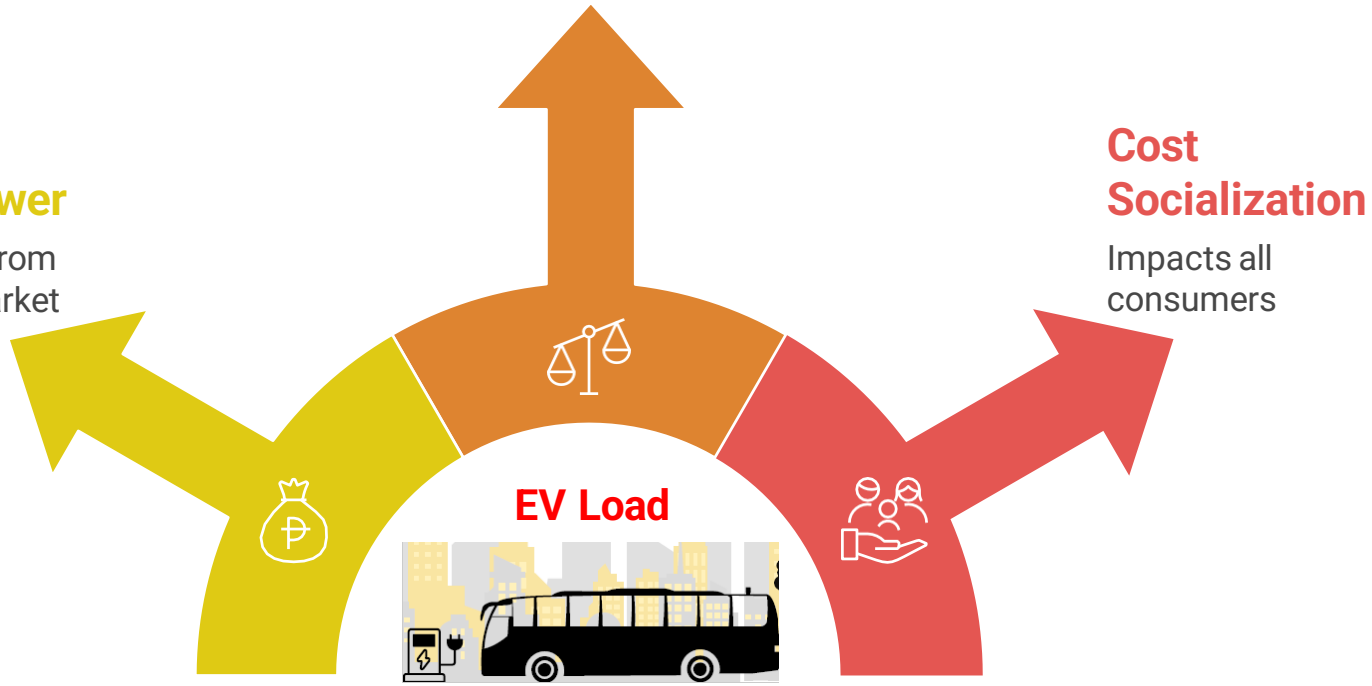
Tariff comparatively higher for other Consumers

Key Challenges with Surging Electric Fleet

KEY CHALLENGES

EV **increases power demand load**, which coincides with **periods of peak demand**, is **intermittent** and **increases power purchase cost** of DISCOMs who have to procure costly power from **short-term market**.

In **Delhi**, the **tariff for EVs is less than 50% of the Average Cost of Supply** (ACoS). In many other states also, the tariff is not in line with MoP rules. This leads to **socialization of cost**, and impacts consumers.



APPROACH G METHODOLOGY

NDA Signed

- NDAs signed with Delhi DISCOMs *(Annexure-2)*
- BSES dtd. 31st Jan'24
- TPDDL gave in Principle approval vide email dtd. 10/01/2024

Scope of Study

- Impact Assessment of EV Load Depot C Power Purchase
- Electric Bus actual 30 mins feeder data of consumption, Power Purchase Quantum C Cost, Revenue details etc.

Data Sharing

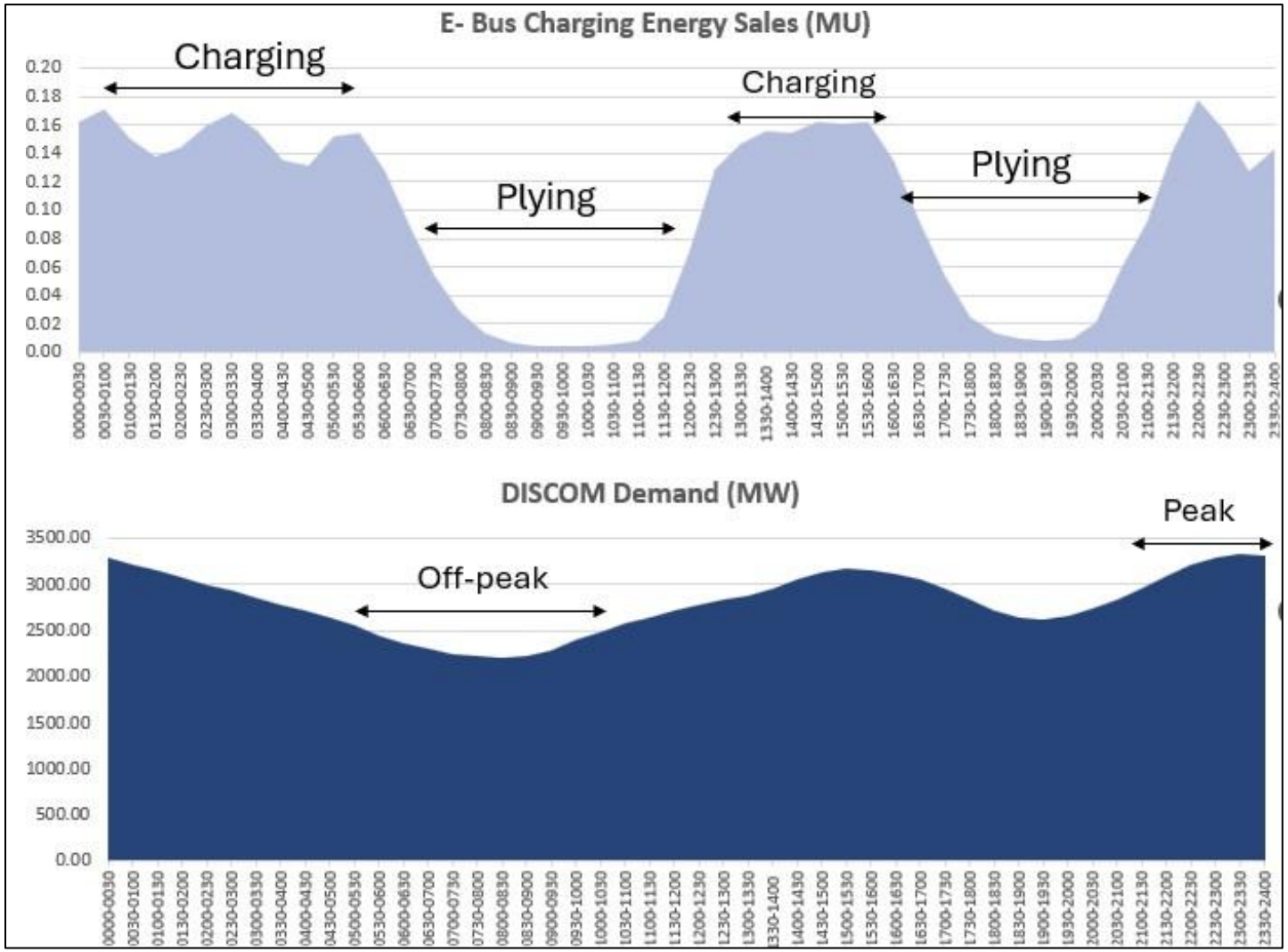
- Actual Feeder information on Half Hourly basis for June, July, Dec 2023 and June, July 2024
- Demand Met of DISCOM as a whole
- Average Loading (kW), Long-term C Short-Term (Power Exchange C Bilateral), Power Purchase Quantum and Cost data

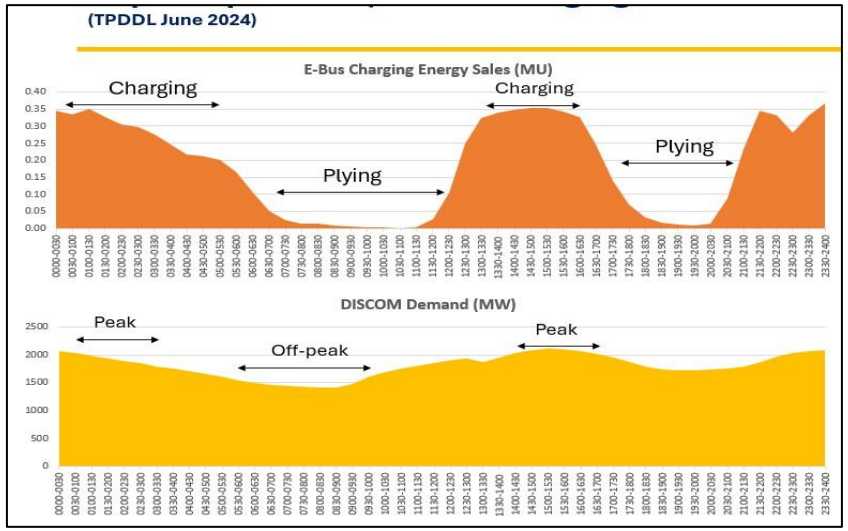
Impact Assessment Study majorly related to Power Purchase Cost, Cross subsidization levels due to Electric Buses Charging Pattern wherein Actual data related to Feeder, catering EV Depot Load, and Power Purchase for Delhi DISCOMs will be utilised.

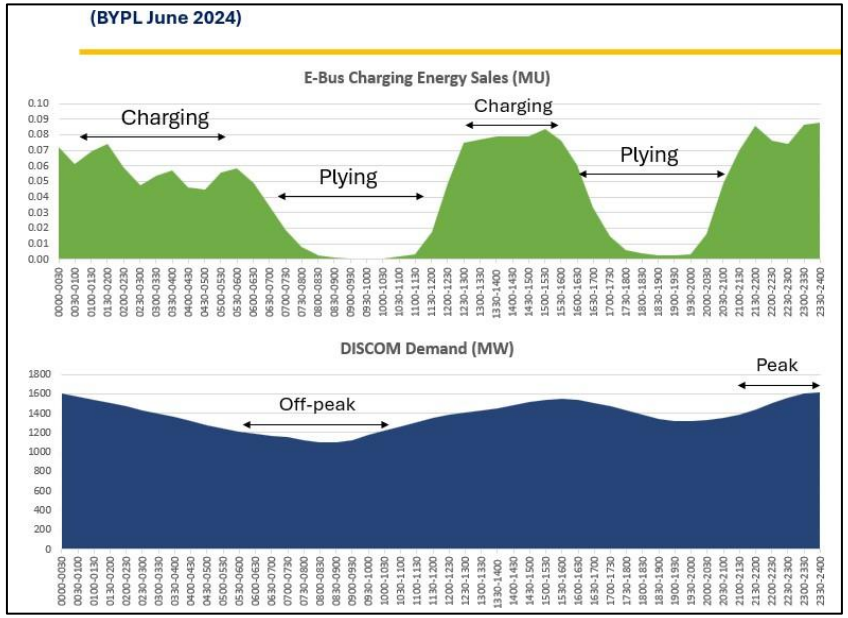
FINDINGS

Superimposition |E-Bus Charging load wrt Total Load

(BRPL June 2024)







Heat Map |E-Bus Charging G total Load in Summer

(BRPL June 2024)

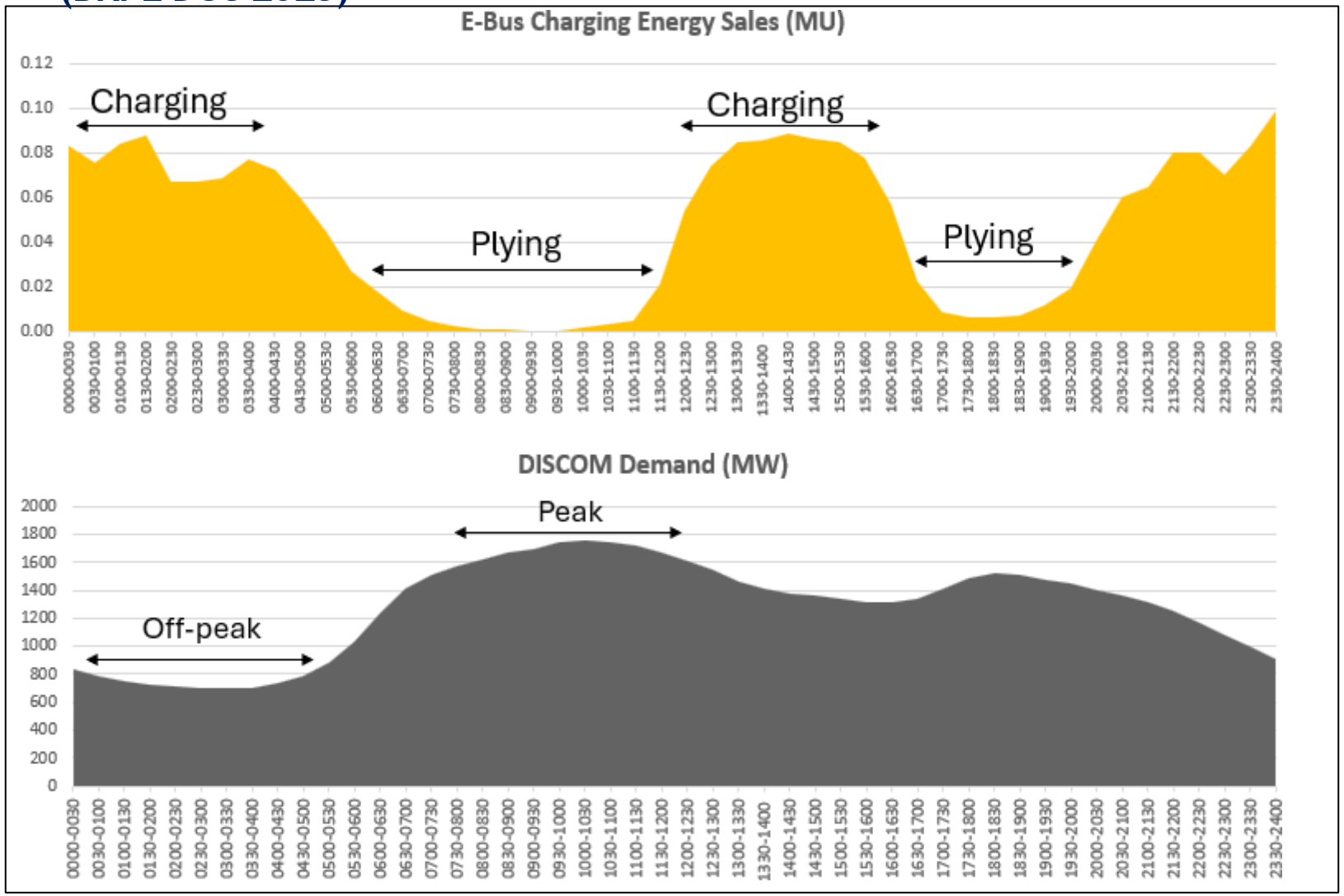
Time Block	Demand Met (MU)	E-Bus Consumption (MU)				Total E-Bus consumption (MU)	Demand met (%) wrt Total Demand Met	E-Bus consumption (%) wrt Total E-Bus
		Mundela (4.8 MW)	Mayapuri (4 MW)	Nehru Place (5 MW)	Sukhdev Vihar (7 MW)			
0000-0030	49.28	0.04725	0.03062	0.02598	0.05866	0.16251	2.44%	3.53%
0030-0100	48.33	0.05043	0.03228	0.03500	0.05290	0.17061	2.39%	3.71%
0100-0130	47.22	0.03639	0.03505	0.03193	0.04690	0.15027	2.34%	3.27%
0130-0200	46.17	0.02479	0.03314	0.02295	0.05713	0.13800	2.29%	3.00%
0200-0230	44.97	0.03078	0.02717	0.02894	0.05678	0.14367	2.23%	3.12%
0230-0300	43.95	0.04273	0.02982	0.03556	0.05135	0.15946	2.18%	3.47%
0300-0330	42.86	0.04675	0.03536	0.02525	0.06085	0.16822	2.12%	3.66%
0330-0400	41.81	0.03852	0.03586	0.02289	0.05800	0.15527	2.07%	3.37%
0400-0430	40.69	0.02979	0.03215	0.02806	0.04467	0.13467	2.01%	2.93%
0430-0500	39.64	0.03215	0.03108	0.02597	0.04260	0.13180	1.96%	2.86%
0500-0530	38.31	0.04136	0.03392	0.01940	0.05706	0.15173	1.90%	3.30%
0530-0600	36.71	0.04314	0.03415	0.01622	0.06068	0.15419	1.82%	3.35%
0600-0630	35.43	0.03938	0.02993	0.01141	0.04667	0.12739	1.75%	2.77%
0630-0700	34.44	0.02890	0.02354	0.00519	0.02978	0.08742	1.71%	1.90%
0700-0730	33.75	0.01712	0.01644	0.00193	0.01878	0.05426	1.67%	1.18%
0730-0800	33.30	0.00917	0.01036	0.00065	0.00864	0.02881	1.65%	0.63%
0800-0830	33.12	0.00485	0.00463	0.00031	0.00408	0.01387	1.64%	0.30%
0830-0900	33.38	0.00256	0.00165	0.00028	0.00292	0.00741	1.65%	0.16%
0900-0930	34.34	0.00136	0.00052	0.00015	0.00260	0.00464	1.70%	0.10%
0930-1000	35.90	0.00097	0.00033	0.00063	0.00245	0.00439	1.78%	0.10%
1000-1030	37.32	0.00075	0.00040	0.00164	0.00208	0.00487	1.85%	0.11%
1030-1100	38.60	0.00077	0.00037	0.00304	0.00174	0.00592	1.91%	0.13%
1100-1130	39.71	0.00079	0.00032	0.00549	0.00170	0.00830	1.97%	0.18%
1130-1200	40.65	0.00732	0.00103	0.01555	0.00161	0.02551	2.01%	0.55%
1200-1230	41.56	0.02909	0.01100	0.01802	0.01279	0.07090	2.06%	1.54%
1230-1300	42.59	0.04049	0.02620	0.01770	0.04390	0.12829	2.11%	2.79%
1300-1330	43.27	0.05084	0.02886	0.02436	0.04332	0.14738	2.14%	3.20%
1330-1400	44.36	0.05439	0.03068	0.02347	0.04782	0.15636	2.20%	3.40%
1400-1430	45.75	0.05143	0.02955	0.02532	0.04869	0.15500	2.27%	3.37%
1430-1500	46.96	0.05504	0.02898	0.02833	0.04940	0.16175	2.33%	3.52%
1500-1530	47.52	0.05252	0.02868	0.02870	0.05049	0.16038	2.35%	3.49%
1530-1600	47.22	0.05618	0.02865	0.02730	0.04948	0.16160	2.34%	3.51%
1600-1630	46.76	0.05268	0.02198	0.02134	0.03889	0.13489	2.32%	2.93%
1630-1700	45.74	0.04326	0.01568	0.01182	0.02226	0.09302	2.26%	2.02%
1700-1730	44.37	0.02614	0.00817	0.00918	0.01101	0.05450	2.20%	1.18%
1730-1800	42.66	0.01030	0.00403	0.00532	0.00594	0.02558	2.11%	0.56%
1800-1830	40.83	0.00431	0.00199	0.00279	0.00413	0.01321	2.02%	0.29%
1830-1900	39.57	0.00295	0.00087	0.00235	0.00340	0.00956	1.96%	0.21%
1900-1930	39.23	0.00201	0.00075	0.00280	0.00242	0.00798	1.94%	0.17%
1930-2000	39.93	0.00161	0.00146	0.00411	0.00233	0.00951	1.98%	0.21%
2000-2030	41.14	0.00157	0.00702	0.01049	0.00197	0.02105	2.04%	0.46%
2030-2100	42.71	0.01276	0.02273	0.02194	0.00183	0.05925	2.11%	1.29%
2100-2130	44.36	0.02896	0.03104	0.02604	0.00557	0.09161	2.20%	1.99%
2130-2200	46.27	0.04982	0.02810	0.02354	0.03991	0.14137	2.29%	3.07%
2200-2230	48.08	0.05250	0.02644	0.02609	0.07217	0.17720	2.38%	3.85%
2230-2300	49.41	0.03573	0.02779	0.03128	0.06251	0.15730	2.45%	3.42%
2300-2330	49.91	0.02229	0.03357	0.03084	0.04034	0.12704	2.47%	2.76%
2330-2400	49.79	0.03105	0.03535	0.02589	0.05090	0.14319	2.47%	3.11%
Total	**201G.87**	**1.35**	**0.G6**	**0.81**	**1.48**	**4.60**	**100.00%**	**100.00%**

During Summers (June 2024) for BRPL:

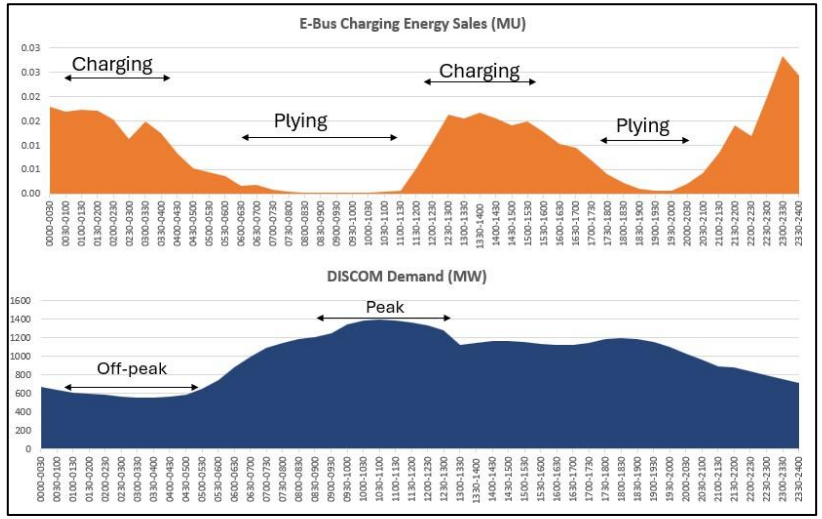
- Peak Load of DISCOM and E-Bus Charging load comes during Night **(10 pm to 2:30 am)** and during afternoon **(2:30 pm to 4 pm)**
- During morning hours (7am to 10am), DISCOM doesn't have Peak Load neither there is load of E-Bus Charging. Similar scenario observed during evening hours (6 pm to 8 pm)

E-Bus Charging load wrt total load in Winter

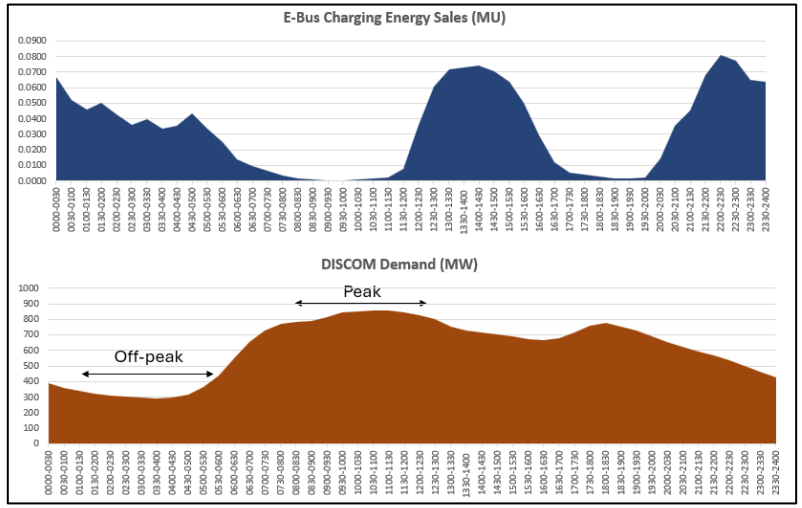
(BRPL Dec 2023)



(TPDDL Dec 2023)



(BYPL Dec 2023)



Heat Map E-Bus Charging G total Load in Winter (TPDDL Dec 2023)

Power Foundation of India

Time Block	Demand Met (MU)	E-Bus Consumption (MU)		Total E-Bus consumption (MU)	Demand Met (%) wrt Total Demand Met	E-Bus consumption (%) wrt Total E-Bus
		Depot 5 -Wazirpur	Depot-6-Subhash Place			
0000-0030	10.07	0.0024	0.0156	0.0179	1.41%	4.23%
0030-0100	9.62	0.0032	0.0139	0.0170	1.35%	4.01%
0100-0130	9.18	0.0037	0.0136	0.0173	1.29%	4.07%
0130-0200	8.94	0.0035	0.0137	0.0171	1.25%	4.04%
0200-0230	8.75	0.0032	0.0122	0.0154	1.22%	3.63%
0230-0300	8.57	0.0026	0.0087	0.0113	1.20%	2.67%
0300-0330	8.38	0.0025	0.0123	0.0149	1.17%	3.50%
0330-0400	8.34	0.0021	0.0105	0.0126	1.17%	2.96%
0400-0430	8.50	0.0012	0.0071	0.0084	1.19%	1.97%
0430-0500	8.86	0.0006	0.0046	0.0052	1.24%	1.23%
0500-0530	9.74	0.0008	0.0037	0.0044	1.36%	1.04%
0530-0600	11.16	0.0010	0.0027	0.0037	1.56%	0.88%
0600-0630	13.22	0.0005	0.0013	0.0018	1.85%	0.41%
0630-0700	15.03	0.0007	0.0012	0.0019	2.11%	0.46%
0700-0730	16.34	0.0005	0.0004	0.0009	2.29%	0.21%
0730-0800	17.15	0.0002	0.0003	0.0005	2.40%	0.12%
0800-0830	17.77	0.0000	0.0003	0.0003	2.49%	0.07%
0830-0900	18.12	0.0000	0.0003	0.0003	2.54%	0.07%
0900-0930	18.82	0.0000	0.0003	0.0003	2.64%	0.06%
0930-1000	20.25	0.0000	0.0003	0.0003	2.84%	0.06%
1000-1030	20.85	0.0001	0.0002	0.0002	2.92%	0.06%
1030-1100	20.99	0.0004	0.0002	0.0006	2.94%	0.13%
1100-1130	20.78	0.0003	0.0005	0.0008	2.91%	0.19%
1130-1200	20.44	0.0003	0.0051	0.0054	2.86%	1.28%
1200-1230	19.95	0.0012	0.0094	0.0106	2.79%	2.50%
1230-1300	19.21	0.0024	0.0139	0.0163	2.69%	3.85%
1300-1330	16.88	0.0029	0.0127	0.0156	2.36%	3.67%
1330-1400	17.13	0.0024	0.0143	0.0167	2.40%	3.94%
1400-1430	17.46	0.0021	0.0133	0.0155	2.45%	3.64%
1430-1500	17.47	0.0018	0.0124	0.0142	2.45%	3.34%
1500-1530	17.26	0.0022	0.0128	0.0150	2.42%	3.54%
1530-1600	17.01	0.0013	0.0113	0.0126	2.38%	2.98%
1600-1630	16.81	0.0012	0.0091	0.0102	2.35%	2.41%
1630-1700	16.88	0.0007	0.0089	0.0096	2.36%	2.26%
1700-1730	17.21	0.0003	0.0066	0.0069	2.41%	1.63%
1730-1800	17.77	0.0001	0.0040	0.0041	2.49%	0.96%
1800-1830	18.00	0.0007	0.0017	0.0024	2.52%	0.56%
1830-1900	17.82	0.0004	0.0007	0.0012	2.50%	0.27%
1900-1930	17.36	0.0001	0.0007	0.0008	2.43%	0.19%
1930-2000	16.58	0.0002	0.0006	0.0008	2.32%	0.18%
2000-2030	15.49	0.0007	0.0014	0.0020	2.17%	0.48%
2030-2100	14.46	0.0003	0.0041	0.0044	2.03%	1.02%
2100-2130	13.46	0.0003	0.0083	0.0086	1.89%	2.03%
2130-2200	13.21	0.0018	0.0124	0.0142	1.85%	3.35%
2200-2230	12.64	0.0019	0.0099	0.0118	1.77%	2.79%
2230-2300	12.04	0.0019	0.0180	0.0199	1.69%	4.69%
2300-2330	11.34	0.0046	0.0237	0.0284	1.59%	6.68%
2330-2400	10.69	0.0043	0.0200	0.0243	1.50%	5.71%
Total	**714.01**	**0.0653**	**0.35G2**	**0.4245**		

FINDINGS |SUMMARIZED

Coinciding E-Bus Charging	Electric Buses Charging **coincides with the Peak Demand of DISCOMs in Summer months** **(April- Sept)**
Power for E-Bus Charging	E-buses Charging load is majorly met through **procurement of Short-term power** at **very high rate** ~8.5 Rs./kWh
Shifting of E-Bus Charging	E-buses Charging load during Peak hours which can **be shifted and met through Long term Power Purchase** subject to availability of Power during that period
BESS for Optimization	Deployment of Battery Storage **(BESS) at Strategic locations** for Power Purchase Cost Optimization of E-Bus Charging
Network Cost	By shifting the Peak Demand occurring through E-Bus Charging Stations, **savings in corresponding Network Cost of the DISCOMs**

1. OPTIMIZATION THROUGH SHIFTING OF E-BUS CHARGING LOAD

OPTIMIZATION OF POWER PURCHASE COST |AVOIDING PEAK | BYPL

Power Foundation of India

June-24 BYPL													
Time Block	Demand Met (MU)	Total E-Bus consumption (MU)	Demand met (%) wrt Total Demand Met	E-Bus consumption (%) wrt Total E-Bus consumption	Long Term (MU)	ST-Exchange (MU) BUY	Net Power Purchase (MU)	LT Price (Rs/kWh)	ST Price (Rs/kWh) BUY	Total available Long Term Power	Scenario-1		
											Existing E-Bus Charging Pattern	New E-Bus Charging	Cost Savings LT
0000-0030	23.99	0.0724	2.42%	3.43%	19.26	0.89	24.75	4.28	10.10		0.07	0.07	
0030-0100	23.57	0.0611	2.38%	2.90%	19.25	0.81	24.32	4.28	10.15		0.06	0.06	
0100-0130	23.10	0.0696	2.33%	3.30%	18.89	0.75	23.74	4.28	10.14		0.07	0.07	
0130-0200	22.61	0.0743	2.28%	3.52%	18.74	0.69	23.26	4.28	9.65		0.07	0.07	
0200-0230	22.07	0.0589	2.23%	2.79%	18.61	0.50	22.70	4.28	8.73		0.06	0.06	
0230-0300	21.51	0.0477	2.17%	2.26%	18.43	0.45	22.18	4.28	9.95		0.05	0.05	
0300-0330	20.95	0.0538	2.11%	2.55%	18.20	0.37	21.63	4.28	8.05		0.05	0.05	
0330-0400	20.40	0.0573	2.06%	2.72%	17.96	0.25	21.08	4.28	6.74		0.06	0.06	
0400-0430	19.74	0.0461	1.99%	2.18%	17.89	0.19	20.50	4.28	7.09		0.05	0.05	
0430-0500	19.21	0.0451	1.94%	2.13%	17.63	0.13	19.96	4.28	6.98		0.05	0.02	0.01
0500-0530	18.63	0.0560	1.88%	2.65%	17.39	0.13	19.28	4.28	9.76		0.06	0.03	0.02
0530-0600	18.08	0.0586	1.83%	2.78%	17.40	0.13	18.71	4.28	9.03		0.06	0.03	0.02
0600-0630	17.78	0.0488	1.79%	2.31%	17.48	0.17	18.41	4.28	9.01	69.90	0.05	0.02	0.02
0630-0700	17.58	0.0342	1.77%	1.62%	17.63	0.15	18.24	4.28	9.27	35.22	0.03	0.09	
0700-0730	17.28	0.0187	1.74%	0.89%	17.59	0.11	17.86	4.28	6.78	**34.68**	0.02	0.07	
0730-0800	16.91	0.0084	1.71%	0.40%	17.73	0.05	17.45	4.28	7.04		0.01	0.01	
0800-0830	16.55	0.0027	1.67%	0.13%	17.79	0.04	17.20	4.28	7.46		0.00	0.00	
0830-0900	16.46	0.0012	1.66%	0.05%	18.17	0.06	17.10	4.28	5.92		0.00	0.00	
0900-0930	16.85	0.0008	1.70%	0.04%	18.54	0.05	17.41	4.28	6.87		0.00	0.00	
0930-1000	17.62	0.0008	1.78%	0.04%	18.97	0.08	18.14	4.28	5.99		0.00	0.00	
1000-1030	18.28	0.0008	1.84%	0.04%	19.28	0.10	18.79	4.28	5.53		0.00	0.00	
1030-1100	19.00	0.0020	1.92%	0.09%	19.51	0.18	19.49	4.28	5.20		0.00	0.00	
1100-1130	19.62	0.0033	1.98%	0.16%	19.58	0.14	20.13	4.28	5.08		0.00	0.00	
1130-1200	20.21	0.0179	2.04%	0.85%	19.54	0.12	20.75	4.28	5.38		0.02	0.02	
1200-1230	20.76	0.0486	2.10%	2.30%	19.70	0.07	21.28	4.28	6.22		0.05	0.05	
1230-1300	21.17	0.0746	2.14%	3.53%	19.82	0.07	21.67	4.28	6.34		0.07	0.07	
1300-1330	21.38	0.0770	2.16%	3.65%	19.90	0.05	21.95	4.28	6.76		0.08	0.08	
1330-1400	21.82	0.0788	2.20%	3.73%	20.06	0.13	22.54	4.28	6.20		0.08	0.08	
1400-1430	22.25	0.0787	2.25%	3.73%	20.45	0.15	23.29	4.28	6.39		0.08	0.08	
1430-1500	22.70	0.0792	2.29%	3.75%	20.77	0.22	24.01	4.28	7.01		0.08	0.08	
1500-1530	23.00	0.0839	2.32%	3.98%	20.99	0.27	24.39	4.28	6.65		0.08	0.08	
1530-1600	23.24	0.0761	2.35%	3.61%	20.82	0.44	24.35	4.28	6.46		0.08	0.08	
1600-1630	23.14	0.0603	2.34%	2.85%	20.38	0.43	23.87	4.28	6.80		0.06	0.06	
1630-1700	22.67	0.0336	2.29%	1.59%	19.96	0.36	23.25	4.28	6.92		0.03	0.03	
1700-1730	22.14	0.0147	2.23%	0.70%	19.37	0.23	22.65	4.28	7.17		0.01	0.01	
1730-1800	21.50	0.0062	2.17%	0.30%	18.75	0.32	22.07	4.28	8.46		0.01	0.01	
1800-1830	20.78	0.0043	2.10%	0.21%	18.59	0.16	21.40	4.28	5.84		0.00	0.00	
1830-1900	20.17	0.0028	2.04%	0.13%	18.59	0.20	20.73	4.28	6.43		0.00	0.00	
1900-1930	19.72	0.0024	1.99%	0.11%	18.86	0.24	20.39	4.28	7.70		0.00	0.00	
1930-2000	19.77	0.0035	1.99%	0.17%	19.06	0.31	20.47	4.28	9.15		0.00	0.00	
2000-2030	19.89	0.0161	2.01%	0.76%	19.35	0.41	20.63	4.28	10.90	**36.48**	0.02	0.09	
2030-2100	20.28	0.0487	2.05%	2.31%	19.32	0.40	20.99	4.28	11.08	38.67	0.05	0.13	
2100-2130	20.76	0.0702	2.09%	3.33%	19.35	0.66	21.69	4.28	11.21	78.14	0.07	0.04	0.04
2130-2200	21.59	0.0857	2.18%	4.06%	19.50	0.93	22.69	4.28	11.34		0.09	0.04	0.05
2200-2230	22.56	0.0763	2.28%	3.62%	19.65	0.89	23.52	4.28	11.32		0.08	0.04	0.04
2230-2300	23.43	0.0742	2.36%	3.51%	19.66	0.73	24.54	4.28	11.37		0.07	0.04	0.04
2300-2330	24.02	0.0866	2.42%	4.10%	19.54	0.88	24.91	4.28	11.40		0.09	0.09	
2330-2400	24.19	0.0881	2.44%	4.17%	19.60	0.92	25.01	4.28	11.10		0.09	0.09	
Total	**GG0.G3**	**2.11**	**100.00%**	**100.00%**	**G13.47**	**16.01**	**1025.34**				**2.10**	**2.10**	**0.23**

E-BUS SCHEDULE |SCOPE FOR SHIFTING

Power Foundation of India

MORNING DUTY NPD					
S. No.	Route	Duty No.	Schedule O/Shedding time	Route time	Schedule I/Shedding time
1	442	1	4:40	7:20	12:00
2	442	2	5:20	7:35	12:55
3	442	3	6:00	7:35	13:35
4	442	4	6:20	7:35	13:55
5	442	5	6:40	7:35	14:15
6	442	6	7:00	7:35	14:35
7	442	7	7:20	7:35	14:55
8	442	8	7:40	7:35	15:15
9	442	9	8:00	7:35	15:35
10	442	10	8:20	7:35	15:55
11	534 A	1	4:15	7:40	11:55
12	534 A	2	4:30	7:40	12:10
13	534 A	4	5:00	7:40	12:40
14	534 A	5	5:15	7:45	13:00
15	534 A	6	5:30	7:40	13:10
16	534 A	7	5:45	7:40	13:25
17	534 A	9	6:15	7:40	13:55
18	534 A	10	6:30	7:40	14:10
19	534 A	12	7:00	7:40	14:40
20	534 A	13	7:15	7:40	14:55
21	534 A	15	7:50	7:20	15:10
22	534 A	17	8:15	7:40	15:55
23	724	23	6:10	7:30	13:40
24	724	24	7:10	7:30	14:40
25	724	25	8:10	7:30	15:40

Bus No. 442 goes out of Depot at 4:40 am and return to Depot at 12pm, remain there for ~2 hrs. Booster charging for **just 1 hr.**

Again Bus No. 442 goes out of Depot at 1:45 pm and return to Depot at 9 pm, remain there for ~8 hrs. **Charge for just 2 hrs.**

Scope for Schedule shifting.

Charging time: ~2 hrs (80% charge in 1 hr)

1ST OUT 1ST IN

Plying starts from 4:40 am till 8 am

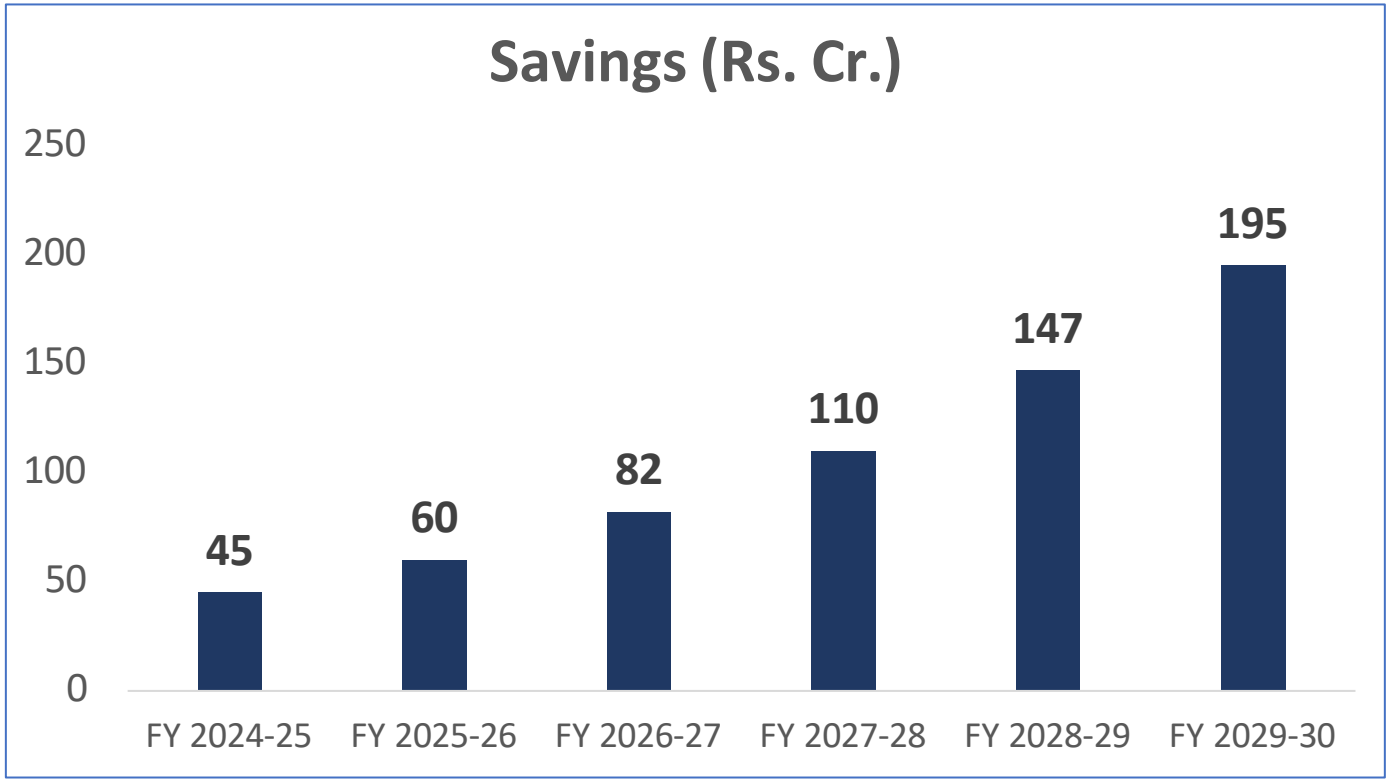
Min 8 hrs running in 1 shift

Booster Afternoon Charging for ~1 hr

Night Charging for 2 hrs.

EVENING DUTY NPD					
S. No.	Route	Duty No.	Schedule O/Shedding time	Route time	Schedule I/Shedding time
1	442	1A	13:45	7:15	21:00
2	442	2A	14:15	7:10	21:25
3	442	3A	14:38	7:37	22:15
4	442	4A	14:58	7:37	22:35
5	442	5A	15:18	7:37	22:55
6	442	6A	15:38	7:37	23:15
7	442	7A	15:58	7:37	23:35
8	442	8A	16:18	7:37	23:55
9	442	9A	16:38	7:37	0:15
10	442	10A	16:58	7:37	0:35
11	534 A	01A	12:55	7:45	20:40
12	534 A	04A	13:40	7:45	21:25
13	534 A	06A	14:10	7:45	21:55
14	534 A	09A	14:55	7:45	22:40
15	534 A	10A	15:10	7:45	22:55
16	534 A	12A	15:40	7:45	23:25
17	534 A	15A	17:00	7:20	0:20
18	534 A	17A	16:40	7:20	0:00
19	724	1A	13:20	7:05	20:25
20	724	24A	15:42	7:33	23:15
21	724	25A	16:45	7:30	0:15

PP Cost Optimization |Shifting of E-Bus Charging Slots



Cumulative savings of

Rs. 638 crore by FY 2029-30

2. OPTIMIZATION THROUGH BATTERY ENERGY STORAGE SYSTEM (BESS)

DUAL PURPOSE -

SAVINGS IN POWER PURCHASE & NETWORK COST

BATTERY ENERGY STORAGE SYSTEM (BESS) |SCOPE

Power Foundation of India

DUAL PURPOSE OF BESS

Energy arbitrage

Energy arbitrage for DISCOMs in optimization of their Power Purchase Cost

Can be charged in off-peak hours and can be discharged in Peak hours

Network Cost Optimization

BESS can be deployed at local network to supply during peak hours to avoid high Power Purchase Cost + Capital Expenditure required to upgrade infra at DISCOMs and Transmission utility end.

BESS: Highly suited for meeting **intermittent Peak Demand of E-Bus Charging** rather than through a tied-up source (PPA)

OPTIMIZATION OF NETWORK + PPC |Funding of BESS through ARR or PSDF

Power Foundation of India

Delhi

FY 2025-26

FY 202G-30

3,035 E-Buses
845 MU

12,000 E-Buses
3,343 MU

PFI Evidence Based Research

All India

1,50,000 E-buses Target by FY 2029-30

Consumption will reach **~42000 MU** by FY 2029-30

Rs. ~13,000 Cr. prospective savings, BESS through PSDF

Scenario 1: PPC saving through shifting

Cumulative Savings till FY 2029-30 (Rs. Cr.)	
Shifting of E-bus charging load	638

Scenario 2: Saving in PPC and Network through BESS (2 Hr storage, 2 cycle) day and night hours

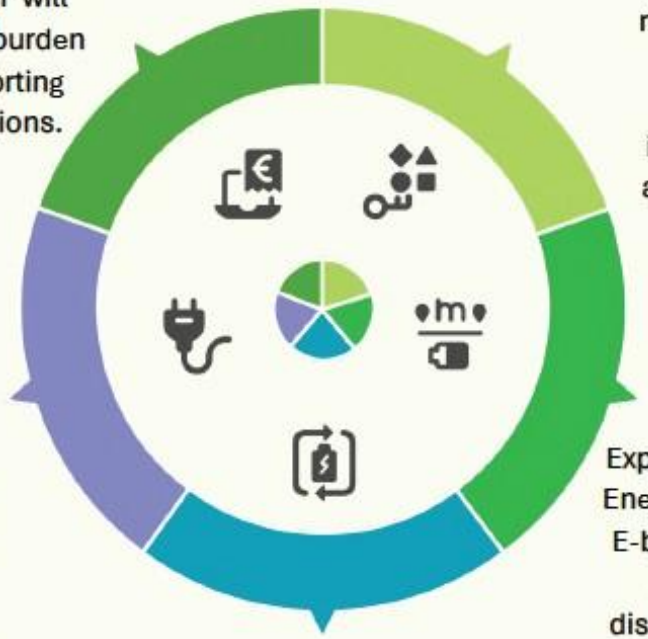
Cumulative Savings till FY 2029-30 (Rs. Cr.)	
BESS – ARR funded	**751**
BESS – PSDF funded **(no CAPEX Burden)**	**1,366**

Scenario 3: Saving in Power Purchase through BESS (4 Hr storage, 1 Cycle) for night hours

Cumulative Savings till FY 2029-30 (Rs. Cr.)	
BESS – ARR funded	**722**
BESS – PSDF funded **(no CAPEX Burden)**	**1,725**

RECOMMENDATIONS TO STAKEHOLDERS

Delhi DISCOMs



Minimise financial burden

Funding BESS through PSDF will help minimise the financial burden on consumers while supporting sustainable system operations.

Resource adequacy plan

Develop a short-term resource adequacy plan to address the growing demand from E-buses, identifying energy storage as the optimal solution for managing this load.

Optimise power purchase

Leverage the PSDF pool to optimise Power Purchase costs, thereby reducing reliance on costly short-term Power procurement.

BESS installation

Explore the installation of Battery Energy Storage Systems (BESS) at E-bus depots, enabling charging during off-peak hours and discharging during peak hours to meet E-bus charging requirements.

Utilising PSDF

Consider utilising the Power System Development Fund (PSDF) to support the deployment of BESS in Delhi for managing intermittent E-bus charging loads.

RECOMMENDATIONS TO STAKEHOLDERS

Delhi Electricity Regulatory Commission (DERC)

Revise the Tariff Schedule for EV charging stations

Tariff of Electric Vehicle Charging Stations should be in line with the provisions of the Tariff Policy, 2016 & Guidelines issued by Ministry of Power (9th September 2024).

Utilise the DERC PSDF Regulation

Support the installation of **Battery Energy Storage Systems (BESS)** at substations for managing intermittent E-bus charging loads. The Delhi PSDF pool currently holds approximately **Rs. 1,445 crore**, contributed by all participating entities, including Delhi DISCOMs.

Leverage PSDF Funding

To avoid loading the Aggregate Revenue Requirement (ARR) with costs of additional network infrastructure. This approach will optimise Power Purchase costs by reducing reliance on expensive short-term procurement or new Power Purchase Agreements (PPAs) to meet short-term load requirements.

RECOMMENDATIONS TO STAKEHOLDERS

Power Foundation of India

STATE GOVERNMENT

DTC may be advised to align the charging schedule of E-Buses with the revised Schedule as recommended in the Study, to optimize on the Power Purchase Cost through shifting of Peak Demand.

Parking Space of DMRC may be utilised to charge E-buses during night hours.

TRANSPORT CORPORATIONS

Utilise the shed area / roof of admin, store C office buildings for installation of Rooftop solar generation.

Adhere to the revised Charging Schedule of E-Buses as worked out in the Study.

Central Assistance (CA) for E-Bus operations per km basis, development/ upgradation of E-Bus Depot, and for Behind the Meter Power Infra (Sub-Station C others)

AGGREGATORS

Aggregators should intimate all future load planning to respective DISCOMs, and **shift charging** from peak hours to normal / off-peak hours to the extent possible

With **coordinated policy, regulatory, and operational interventions**, India can achieve its EV adoption goals while ensuring that the **power sector remains cost-effective and resilient, without increasing ACS much.**

Power Foundation of India

THANK YOU !

OPTIMIZATION OF NETWORK COST |ASSUMPTIONS

Sl. No.	Particulars	UoM	FY 26	FY 27	FY 28	FY 2G	FY 30
1	E-Bus Charging Capacity that can be avoided	MW	**24**	**10**	**14**	**20**	**28**
		MVA*	43	18	25	35	49
2	Capex Requirement for the network upgradation	Rs Cr/MVA	4	4	4	4	4
		Rs Cr	171	70	99	140	197
3	Capex Deferral	Rs Cr	171	70	99	140	197
4	Debt : Equity		70:30	70:30	70:30	70:30	70:30
5	Debt	Rs Cr	120	49	69	98	138
6	Interest Rate on Debt		12%	12%	12%	12%	12%
7	Useful life of the network	yrs	25	25	25	25	25
8	Loan repayment period	yrs	12	12	12	12	12
9	Equity	Rs Cr	51	21	30	42	59
10	Rate of RoCE		14.00%	14.00%	14.00%	14.00%	14.00%
11	Salvage value of Assets		10%	10%	10%	10%	10%
12	Depreciation rate	1st 12 years	5.83%	5.83%	5.83%	5.83%	5.83%
		for next 13 years	1.54%	1.54%	1.54%	1.54%	1.54%
13	Discount rate	9.71%					

- MW converted to MVA considering (n-1) impact with 1.75 redundancy
- CAPEX requirement for the Network Rs. 4 Cr./MVA, as provided by DISCOMs
- Impact of Network Cost computed using **applicable Regulatory provisions (DERC Regulations)**
- Depreciation C return on Capital Employed computed for 25 years
- Weighted Average Cost of Capital (WACC) as

$$WACC = \left[\frac{D}{D+E}\right] * r_d + \left[\frac{E}{D+E}\right] * r_e$$

Where, D= Debt, E=Equity

Re C Rd as Cost of Debt C Equity resp.

Power Foundation of India

SESSION 6:

ELECTRIC VEHICLES – ELECTRIC VEHICLES – A $200 BILLION OPPORTUNITY IN INDIA TO BE UNLOCKED

Session Partner: Power Foundation of India (PFI)

13th March 2026

GOVERNMENT INITIATIVES

National Electric Mobility Mission Plan (NEMMP 2020)

In 2013, the Government of India **launched the National Electric Mobility Mission Plan 2020** with an aim to **achieve national fuel security** by promoting hybrid and electric vehicles in the country.

Targets under NEMMP: 6–7 million hybrid and electric vehicles on Indian roads by 2020.

NEMMP uses different policy-levers such as:

1. Demand side incentives to facilitate acquisition of hybrid/electric vehicles
2. Promoting RCD in technology including battery technology
3. Promoting charging infrastructure
4. Supply side incentives
5. Encouraging retro-fitment of on-road vehicles with hybrid kit

Faster Adoption and Manufacturing of Electric (GHybrid) Vehicles in India (FAME)

As part of the NEMMP 2020, the Ministry of Heavy Industries formulated the FAME scheme in March 2015 with an aim to reduce dependency on fossil fuel and to address issues of vehicular emissions.

Scheme Period: FAME-I (2015 - 19); FAME-II (2019-24)
Total outlay: FAME-I (Rs. 895 Cr.); FAME-II (Rs 11,500 Cr.)

- FAME-I had four focus areas - Technological development, Demand generation, Pilot project and Charging infrastructure components
- Key elements of Phase-II included Demand incentives for EVs; Charging infrastructure development; and phased manufacturing programme for localisation of components

Achievements: ~2.5 lakh EVs supported; and ~520 charging stations sanctioned under Phase I.

Under Phase II, ~16.3 lakh EVs supported; and ~9,332 charging stations sanctioned. (as of June 2025)